

18

A

From single slit diffraction formula, $\sin \theta = \lambda / b$

or $x / D = \lambda / b$ where D is the distance from slit to screen

B, C and D are correct as seen from the equation. However, when slit width is increased, more light passes through the slit and the intensity increases. Hence the area under the graph increases.

19

D

Time for electron to travel in electric field, $t = \frac{S_x}{v_x}$